

# Optimization-Based Clinical Decision Support for Curable Hypertension

**SVMs: Hard-Margin Feasibility, Soft Margins, and Kernel Validation**

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# The story in one line

strict  $\rightarrow$  prove impossible  $\rightarrow$  relax  $\rightarrow$  compare  $\rightarrow$  conclude

- Try the **hard-margin** SVM: a straight line with zero mistakes.
- **Prove** no such line exists — three independent ways.
- **Relax** to a soft margin that charges for mistakes.
- **Compare** against a curved (RBF kernel) boundary.
- **Conclude**: the simple linear model is the right choice.

# The five steps — what each one asks and answers

- 1 Hard margin — can a perfect line exist?** Demand a straight boundary that classifies every patient correctly with a margin. This is a convex feasibility problem: either such a line exists, or it provably does not.
- 2 Prove it is impossible.** Three independent checks agree the classes are not linearly separable: (2a) the LP solver returns “infeasible”; (2b) a 12-patient Farkas certificate turns that into a proof ( $0 \geq 1$ ); (2c) the two convex hulls overlap.
- 3 Relax to the soft margin.** Allow mistakes but charge for them through slack variables and a penalty  $C$ . Always solvable; (3a) tune  $C$  by grid search, (3b) read it as a Pareto trade-off, (3c) see the fitted line, (3d) solve the primal directly by SGD.
- 4 Compare against a curved boundary.** Swap the inner product for an RBF kernel so the boundary can bend, then check whether the extra flexibility actually helps or merely overfits.

**Takeaway:** Steps 1–2 establish *what is impossible*; steps 3–4 find *the best thing that is possible*.

# The clinical problem

**Primary aldosteronism (PA)** is a common, *curable* cause of high blood pressure.

- **Unilateral (UPA)**: one gland — often *cured by surgery*.
- **Bilateral (BPA)**: both glands — managed with *medication*.

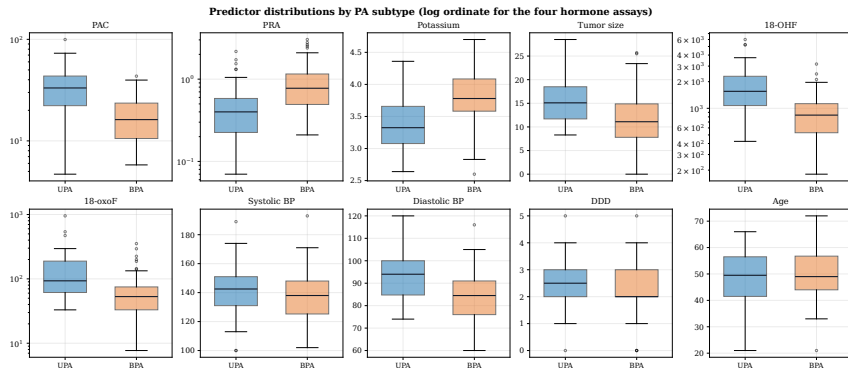
Getting the subtype right changes the treatment.  
Can an optimization model separate the two from routine measurements?

**Takeaway:** Frame subtype classification as a maximum-margin optimization problem.

## The dataset

- 114 synthetic patients
- 56 UPA / 58 BPA
- 10 continuous predictors (PAC, PRA, potassium, tumor size, steroids, BP, ...)

# What the data look like before any model



114 patients, no missing values. **UPA** versus **BPA**; hormone assays on a log ordinate.

**Takeaway:** Every panel overlaps — no single measurement splits the two subtypes.

# Descriptive statistics — the subtype contrast

Lecture ref: *Statistical Learning Models (W9)*

| Feature             | UPA   | BPA   | Cohen's $d$  | Welch $t$    | $p$              |
|---------------------|-------|-------|--------------|--------------|------------------|
| PAC (ng/dL)         | 34.9  | 18.4  | <b>1.17</b>  | <b>6.20</b>  | <b>&lt;0.001</b> |
| 18-OHF (ng/dL)      | 1883  | 942   | <b>0.99</b>  | <b>5.25</b>  | <b>&lt;0.001</b> |
| Potassium (mmol/L)  | 3.38  | 3.79  | <b>-0.95</b> | <b>-5.09</b> | <b>&lt;0.001</b> |
| PRA (ng/mL/h)       | 0.51  | 0.96  | <b>-0.80</b> | <b>-4.32</b> | <b>&lt;0.001</b> |
| Diastolic BP (mmHg) | 92.3  | 84.1  | <b>0.75</b>  | <b>4.03</b>  | <b>&lt;0.001</b> |
| Tumor size (mm)     | 15.4  | 11.4  | <b>0.74</b>  | <b>3.98</b>  | <b>&lt;0.001</b> |
| 18-oxoF (ng/dL)     | 148.1 | 71.8  | <b>0.66</b>  | <b>3.46</b>  | <b>&lt;0.001</b> |
| DDD (drugs)         | 2.48  | 2.14  | 0.36         | 1.94         | 0.055            |
| Systolic BP (mmHg)  | 141.4 | 137.5 | 0.23         | 1.22         | 0.224            |
| Age (years)         | 48.6  | 50.1  | -0.15        | -0.80        | 0.425            |

Bold:  $p < 0.05$  (Welch  $t$ -test), i.e. separates the subtypes (7 of 10). Sign: + means higher in UPA.

## Cohen's $d$ — how big

Gap between *class means*, in pooled SDs: 0.2 small, 0.5 medium, 0.8+ large.

## Welch $t$ — how certain

Same gap  $\div$  standard error; grows with  $\sqrt{n}$ . Evidence, not size.

**Takeaway:** Strong univariate signal, but overlapping distributions — a perfect separator is in doubt.

# Step 1 — the strictest model: hard-margin SVM

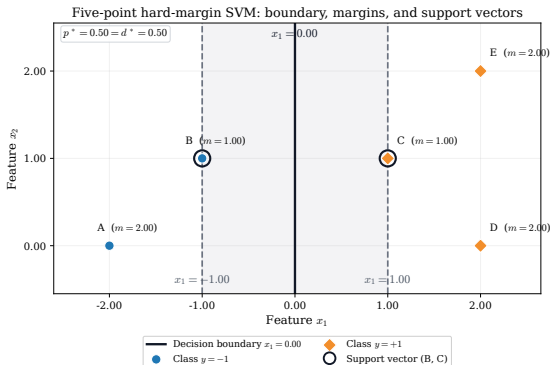
Find the **widest empty corridor** between the two classes:

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad \text{s.t.} \quad t_i(w^\top x_i + b) \geq 1.$$

- Score  $s = w^\top x + b$ ; sign is the prediction.
- Margin width  $= 2/\|w\|$ .
- **Zero mistakes allowed.**

A convex quadratic program — *if* a feasible line exists.

Lecture ref: *Statistical Learning Models (W9)*; *Convex Optimization Models (W8)*



Five-point teaching example: boundary, margins, support vectors.

## Step 2 — prove no such line exists

Lecture ref: *Introduction to Duality (W9); First-Order Optimality (W10)*

We attack the question **three independent ways**. If any one succeeds, the hard-margin model is impossible — and all three agree.

### 2a. Ask the solver

Run the feasibility LP on all data and every fold.

### 2b. Farkas certificate

Get a signed receipt — 12 patients that *prove*  $0 \geq 1$ .

### 2c. Convex hulls

Show the two classes literally share a point.

Each is stronger than the last: a solver *says* no; Farkas *proves* no; the hulls *show* why.



## Step 2a — ask the solver

Feasibility of the hard-margin constraints is itself a **linear program**: does any  $(w, b)$  satisfy

$$t_i(w^\top x_i + b) \geq 1 \quad \text{for all } i?$$

- HiGHS returns **status: infeasible** on the full 114-patient dataset.
- Repeated inside every one of the 5 cross-validation folds — **all infeasible**.
- So the result is not an artifact of one particular split.

**Takeaway:** A “no” from the solver is a starting point — but why should we trust it? Steps 2b–2c give proof.

### Solver verdict

| Subset          | Feasible? |
|-----------------|-----------|
| Full data (114) | No        |
| Fold 1          | No        |
| Fold 2          | No        |
| Fold 3          | No        |
| Fold 4          | No        |
| Fold 5          | No        |

## Step 2b — the Farkas certificate

**Farkas' lemma:** exactly *one* of two systems can hold — either a separating  $w$  exists, **or** a set of nonnegative weights  $\lambda$  gives a contradiction.

- The solver hands back **12 patients** with weights  $\lambda_i \geq 0$ .
- They combine to prove  $0 \geq 1$  — impossible.
- This is a *checkable receipt*: no need to trust the solver's search.

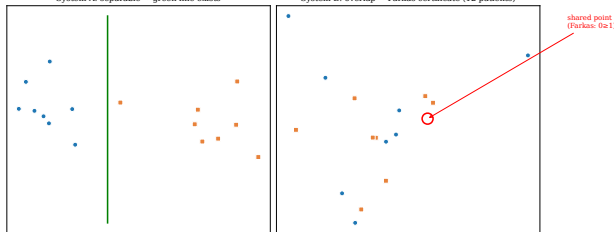
**Takeaway:** The certificate turns “the solver said no” into a proof anyone can verify by hand.

Lecture ref: *Introduction to Duality (W9) — theorem of the alternative*

**Farkas' lemma: exactly one system is solvable**

System A: separable → green line exists

System B: overlap → Farkas certificate (12 patients)



Only one column can be true. On this data it is the right one.

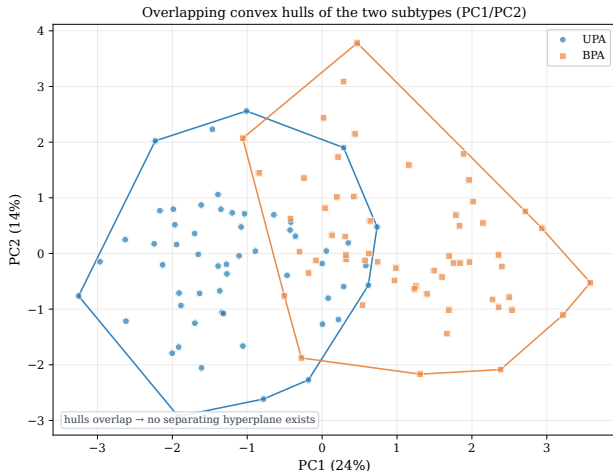
## Step 2c — what the certificate means: overlapping hulls

The same  $\lambda$  weights have a picture.  
Rescaled, they build **one point inside both classes' convex hulls**:

$$\sum_i \beta_i^+ x_i = \sum_i \beta_i^- x_i,$$

matching to  $6.66 \times 10^{-16}$  (machine zero).

- A shared point  $\Rightarrow$  no hyperplane separates the classes.
- Geometry, solver, and algebra tell one story.



The real cohort in a 2-D projection: the hulls clearly overlap. The full-dimensional proof is exact.

# Step 3 — relax: the soft-margin SVM

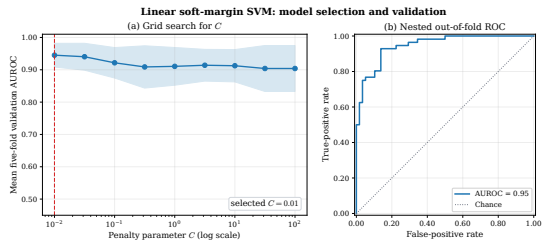
Allow mistakes, but **charge** for them with slack  $\xi_i$ :

$$\min_{w,b,\xi} \quad \frac{1}{2} \|w\|^2 + C \sum_i \xi_i$$

$$\text{s.t.} \quad t_i(w^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0.$$

- $C$  = strictness knob (chosen by nested CV).
- Selected  $C = 0.01$  (leakage-safe grid search).
- Always solvable — overlap is modeled, not assumed away.

Lecture ref: *Statistical Learning Models (W9)* — hinge loss, slack, penalty  $C$



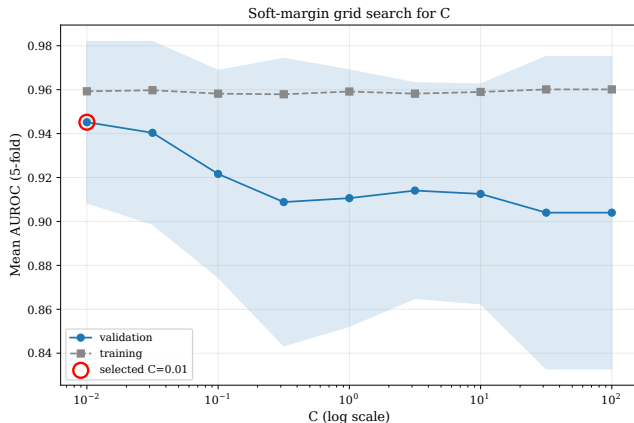
(a) validation AUROC vs  $C$ ; (b) nested out-of-fold ROC, **AUROC 0.95**.

## Step 3a — choosing $C$ by grid search

Sweep  $C$  over  $10^{-2} \dots 10^2$  and score each by **inner-fold validation** AUROC (never the test fold — leakage-safe).

- Training AUROC stays flat near 0.96 at every  $C$ .
- Validation AUROC *peaks at the smallest*  $C = 0.01$ .
- Larger  $C$  (harder penalty)  $\Rightarrow$  worse generalization.

The gap between the two curves *is* the overfitting.



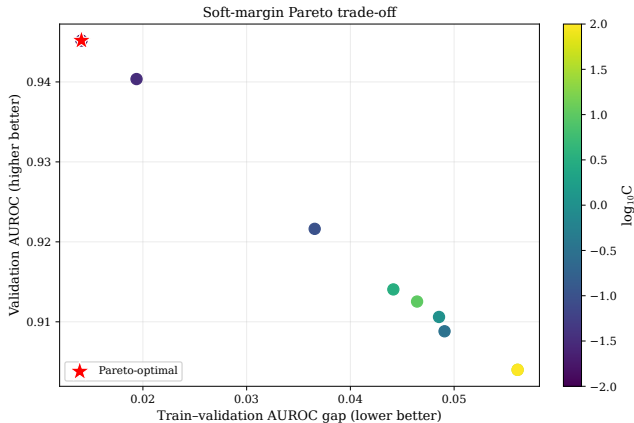
Mean validation vs training AUROC across the  $C$  grid; band is  $\pm 1$  SD over folds. The selected  $C = 0.01$  is circled.

**Takeaway:** Stronger regularization (small  $C$ ) generalizes best on this overlapping data.

## Step 3b — the same grid as a Pareto trade-off

Plot each  $C$  as (train-validation gap, validation AUROC). The best model is **top-left**: high AUROC *and* little overfitting.

- $C = 0.01$  is the **single Pareto-optimal** point — it wins on *both* axes at once.
- Every larger  $C$  gives away AUROC *and* widens the gap.
- No trade-off to negotiate: the simplest penalty is strictly best here.



Color encodes  $\log_{10} C$ ; the star is the selected, Pareto-optimal model.

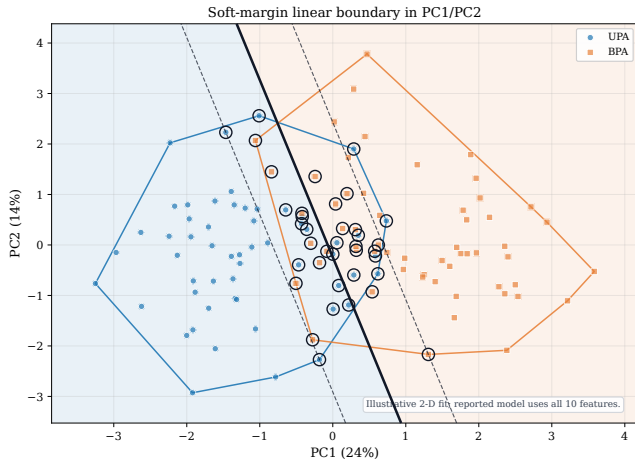
**Takeaway:**  $C = 0.01$  dominates the grid on both axes at once.

## Step 3c — what the soft-margin line looks like

Projecting the cohort onto its first two principal components makes the fitted **soft-margin boundary** visible.

- The line runs *straight through the overlap* where the two hulls intersect.
- Circled points are **support vectors** — almost all sit in the contested band.
- Points inside the margin or on the wrong side are exactly what the slack  $\xi_i$  pays for.

This 2-D fit is illustrative; the reported model uses all 10 features.

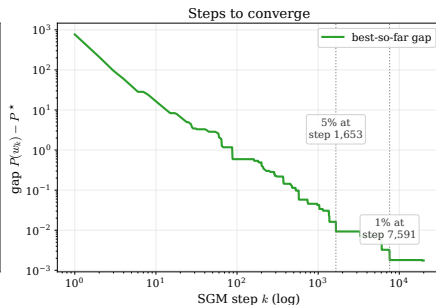
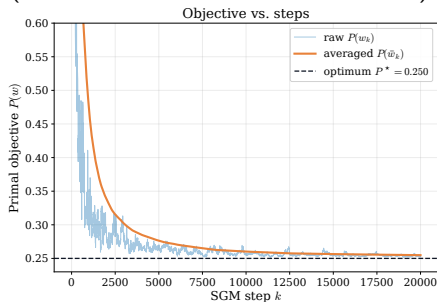


Solid: decision boundary; dashed: the  $\pm 1$  margins. Hulls overlap, so no line separates them cleanly.

## Step 3d — reaching the same solution with SGD

Lecture ref: *The Stochastic Gradient Method; Strong Convexity & Descent (W10)*

The soft-margin objective is a **finite sum** over the  $m$  patients. This means we can optimize it one randomly selected patient at a time, without solving the dual QP:  $w_{k+1} = w_k - \eta_k g_k$  with  $g_k = \lambda w_k$  if  $t_k w_k^\top x_k \geq 1$ , else  $g_k = \lambda w_k - t_k x_k$  (step  $\eta_k = 1/(\lambda k)$ ; at the hinge, we use a *subgradient* together with the regularization term). It comes within 5% of the optimum after about 14 epochs and within 1% after about 67. Its final direction closely matches the dual solution (cosine 0.99; AUROC 0.958 vs 0.955).



**Takeaway:** SGD gets very close to the dual solution. The remaining fluctuations reflect stochastic optimization, not a new test of separability.



## Step 4 — from soft margin to a kernel

Lecture ref: *Introduction to Duality (W9); Statistical Learning Models (W9) — kernels*

### 1. The soft-margin primal (what we solved in Step 3):

$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_i \xi_i \quad \text{s.t.} \quad t_i(w^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0.$$

### 2. Its Lagrangian dual — the data enters **only through inner products** $x_i^\top x_j$ :

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j t_i t_j \underbrace{x_i^\top x_j}_{\text{inner product}} \quad \text{s.t.} \quad 0 \leq \alpha_i \leq C, \quad \sum_i \alpha_i t_i = 0.$$

### 3. The kernel trick. Map $x \mapsto \varphi(x)$ into a higher space and replace every inner product by a **kernel**:

$$x_i^\top x_j \longrightarrow K(x_i, x_j) = \varphi(x_i)^\top \varphi(x_j).$$

### 4. Choose the RBF kernel:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2).$$

A curved boundary in the original space, still a convex QP in  $\alpha$ .

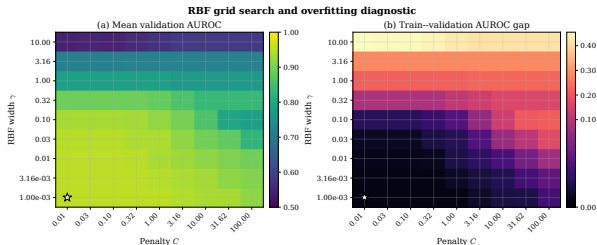
$$\text{Decision score becomes } f(x) = \sum_i \alpha_i t_i K(x_i, x) + b.$$

## Step 4 — compare against a curved boundary (RBF kernel)

The RBF kernel  $K(x_i, x_j) = e^{-\gamma \|x_i - x_j\|^2}$  can bend the boundary.

- Joint search over 81  $(C, \gamma)$  pairs, nested CV.
- Best:  $C = 0.01$ ,  $\gamma = 0.001$  — the *smoothest* setting.
- Large  $\gamma$ : perfect training, near-chance validation (overfit).

The data wanted a nearly-straight boundary.

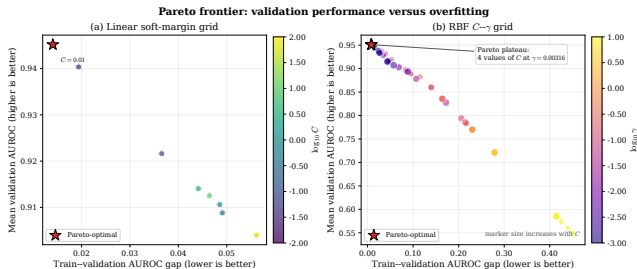


Validation AUROC (left) vs train-validation gap (right). The star marks the selected smooth model; the top rows overfit.

## Step 4 — performance versus overfitting: the Pareto view

Plot every candidate as (train-validation gap, validation AUROC). The **best models sit in the top-left**: high AUROC, small gap.

- **Linear**: the whole frontier is tightly clustered top-left — little room to overfit.
- **RBF**: a long tail trades generalization for training fit as  $\gamma$  grows.
- The Pareto-optimal RBF point coincides with the *smoothest* setting — essentially the linear model.



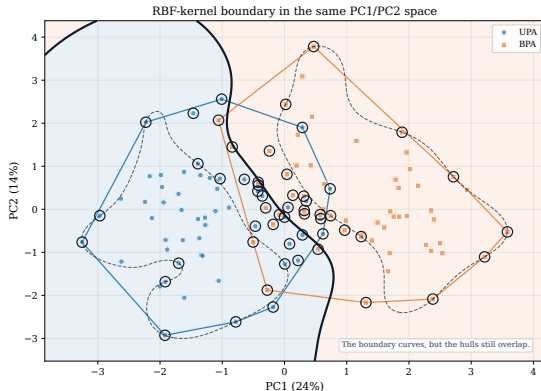
The star is the Pareto-optimal model in each grid. RBF's extra flexibility only buys a longer overfitting tail, not a better frontier.

## Step 4 — the kernel boundary redrawn in PC1/PC2

The same projection as Step 3c, now with the **RBF-kernel** decision surface.

- The boundary is **curved**, not straight.
- It bends *around* points, yet the two hulls still overlap.
- Curvature relabels a few borderline points but does not separate the classes.

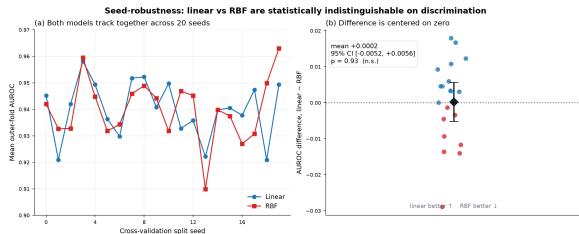
Illustrative 2-D fit (moderate  $\gamma$ ); the tuned model chose a much smoother  $\gamma$ .



Compare with the straight line of Step 3c: more shape, same overlap.

**Takeaway:** The kernel can bend the boundary — but bending does not make overlapping classes separable.

# Does the curved boundary actually help?



We repeated nested cross-validation with 20 different fold assignments and compared mean outer-fold AUROC:

- Difference = +0.0002
- 95% CI [-0.005, +0.006]
- Paired  $t$ -test:  $p = 0.93$
- Linear matched or beat RBF in 12 of 20 runs

In practical terms, the two models discriminate equally well.

**Takeaway:** Across repeated splits, the RBF model shows no reliable advantage over the simpler linear model.

## Step 5 — conclude

### Verdict

- Hard-margin → **impossible** (proven three ways).
- Linear soft-margin → **AUROC 0.95** ✓
- RBF kernel → AUROC 0.94 (no gain).

### The simpler model wins.

Fewer tuning knobs, an interpretable score, equal discrimination — parsimony decides.

### Why it matters

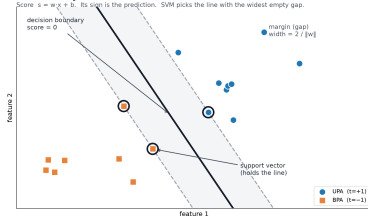
- The negative feasibility result is *informative*, not a bug: clinical measurements genuinely overlap.
- Complexity must be justified by *validation*, never by training fit.
- Synthetic data  $\Rightarrow$  methodological, not clinical, claim.

## Support Vector Machines — One-Page Cheat-Sheet

*strict* → *prove impossible* → *relax* → *compare* → *conclude*

### A. Anatomy: what an SVM draws

Score  $s = w \cdot x + b$ . Its sign is the prediction. SVM picks the line with the widest empty gap.



### B. The report's logic, step by step

#### 1. Try the STRICTEST model

Hard-margin: a straight line with ZERO mistakes allowed.

#### 2. PROVE it's impossible

3 ways agree: solver says 'infeasible', a 12-patient Farfars receipt gives  $0 \geq 1$ , and the two groups overlap exactly.

#### 3. RELAX just enough

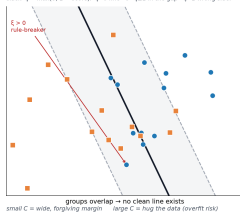
Soft-margin: allow mistakes, but charge for them (knob C). Always solvable. → held-out AUROC = 0.95 ✓ winner

#### 4. COMPARE vs. curved

RBF kernel (knob  $\gamma$ ). Tuning chose the smoothest  $\gamma$ ; AUROC 0.94 — no better. High  $\gamma$  overfits badly: → keep it simple.

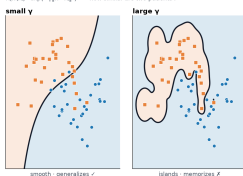
### C. Soft-margin: mistakes cost money

slack  $\xi = \max(0, 1 - \text{score})$ .  $\xi=0$  fine.  $0 < \xi \leq 1$  in the gap.  $\xi > 1$  wrong side.



### D. The $\gamma$ knob (RBF): smooth vs. memorize

$K(x, x) = \exp(-\gamma \|x - x_0\|^2)$  — 'how similar are two patients'.



Here tuning chose the smoothest  $\gamma$  — the data wanted a nearly-straight boundary.

### E. Symbol key & verdict

|                 |                                     |
|-----------------|-------------------------------------|
| $x$             | the 10 measurements for a patient   |
| $t = \pm 1$     | the answer: +1 UPA, -1 BPA          |
| $w$             | direction of the dividing line      |
| $b$             | offset — shifts the line            |
| $w \cdot x + b$ | the score (sign = prediction)       |
| $\ w\ $         | length of $w$ : shorter = wider gap |
| $a$             | how hard a patient pulls the line   |
| $\xi$           | a patient's rule-breaking amount    |
| $C$             | strictness knob for mistakes        |
| $\gamma$        | curviness knob (RBF only)           |

#### VERDICT

Hard-margin → impossible (proven)

Linear soft-margin → AUROC 0.95 ✓

RBF kernel → AUROC 0.94 (no gain)

Simpler model wins.

# Thank you

Questions?

Reproduced end-to-end: 30/30 result files match; independent nested-CV and statistical tests confirm the verdict.